

Date: 2026/6/08

Computer Networks

LAN- Assignment 5

Topics :

- Introduction to LANs
- LAN Protocols
- MAC and LLC
- Classification of MAC
- Ethernet LAN
- IEEE 802.3 , 802.5 , 802.11

Present by :

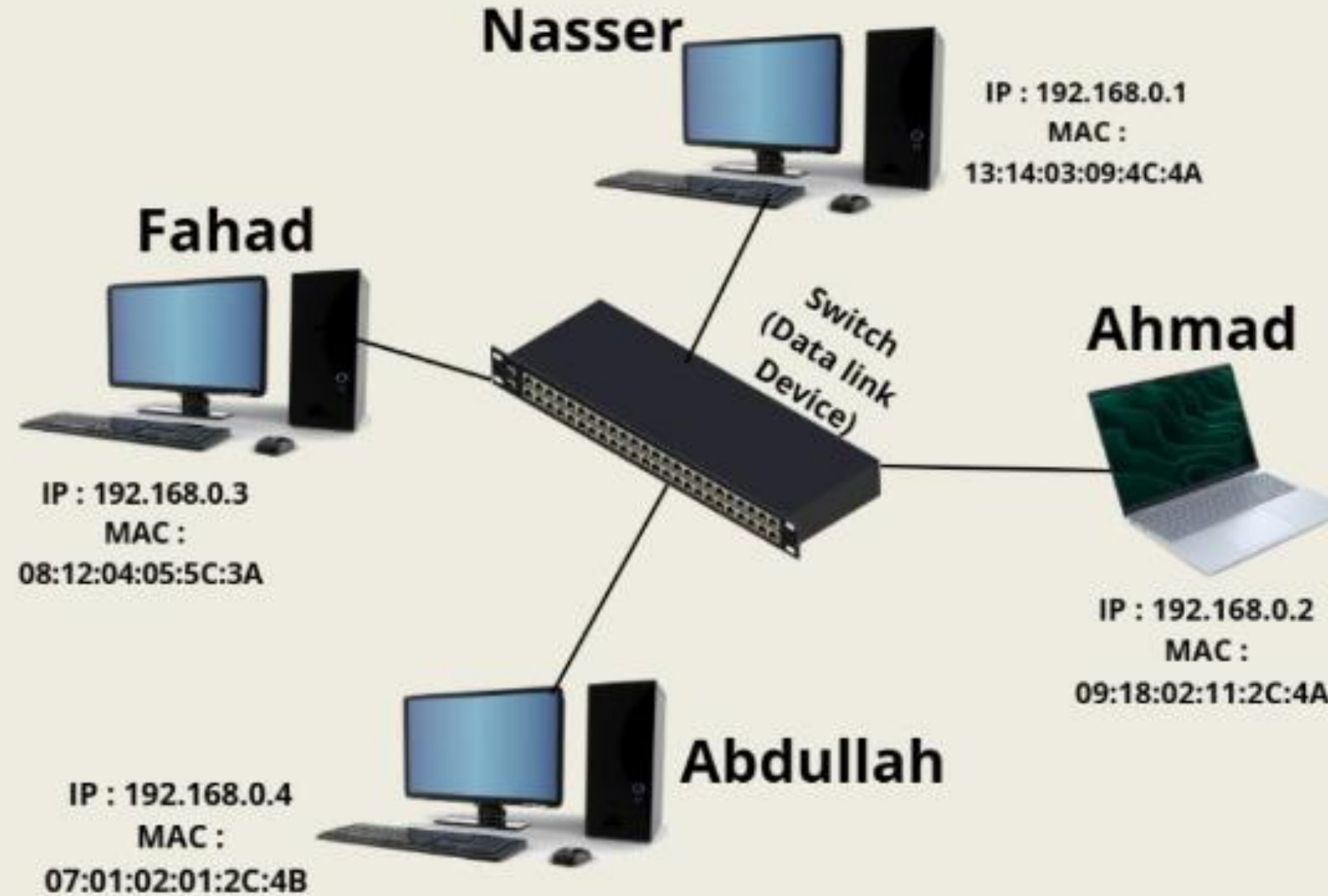
Saad Almalki/0xSaad

Introduction

LAN or Local Area Network is Connection between computer and devices in limited area .

local area network allows multiple nearby computers to use shared resources and communication.

Introduction



LAN

LAN Protocols

Protocols designed for LANs are normally concerned with the data-link layer and the physical layer.

LAN Protocols in :

Data Link

Physical

LAN Protocols

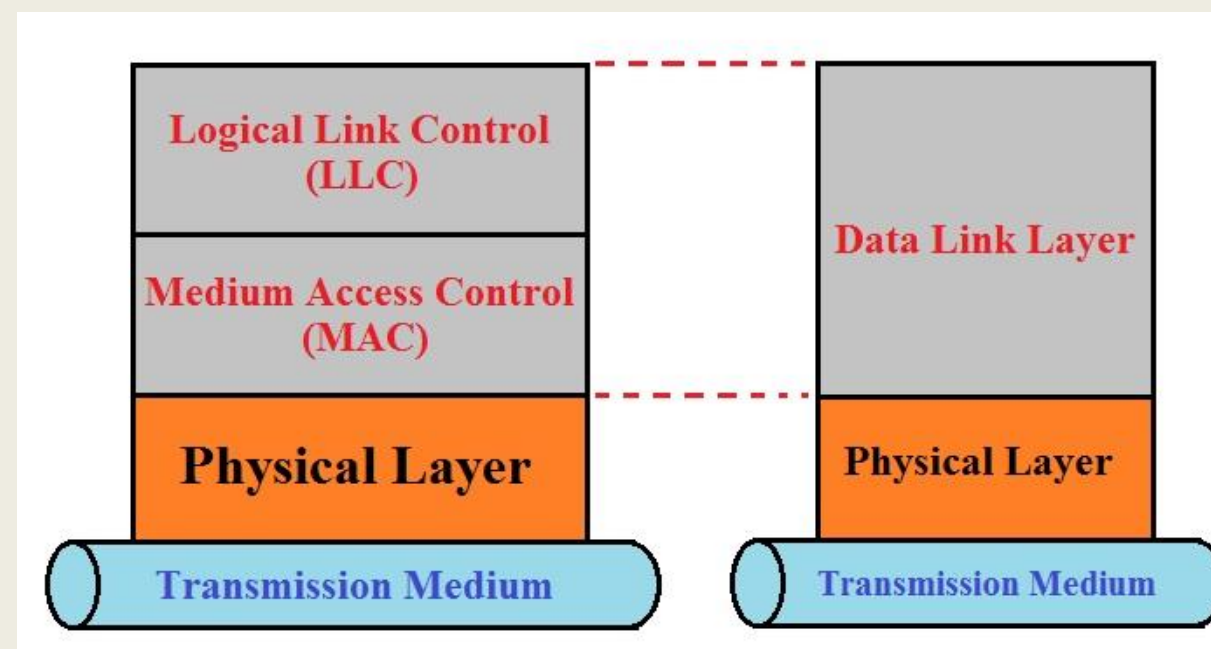
The physical layer of a LAN implements such functions as :

- signal transmission and receiving
- encoding and decoding
- generation and removal of synchronization information
- specifies the type of medium used for transmission
- Specifies the network topology.

LAN Protocols

The IEEE 802 Standards subdividing data-link layer into two main components :

- Logical Link Layer (LLC)
- Medium Access Control (MAC)



MAC Address

MAC (Medium Access Control) : or Physical Address. It's 6 Byte globally unique ID assigned by the manufacturer

MAC address is suitable when multiple devices are connected with same physical link

Example :

MAC Address: f8:ff:c2:4d:32:bd

LLC

LLC or Logical Link layer is sublayer in Data-link layer implements flow and error control and acts as an interface between the (MAC) sublayer and the Network layer .

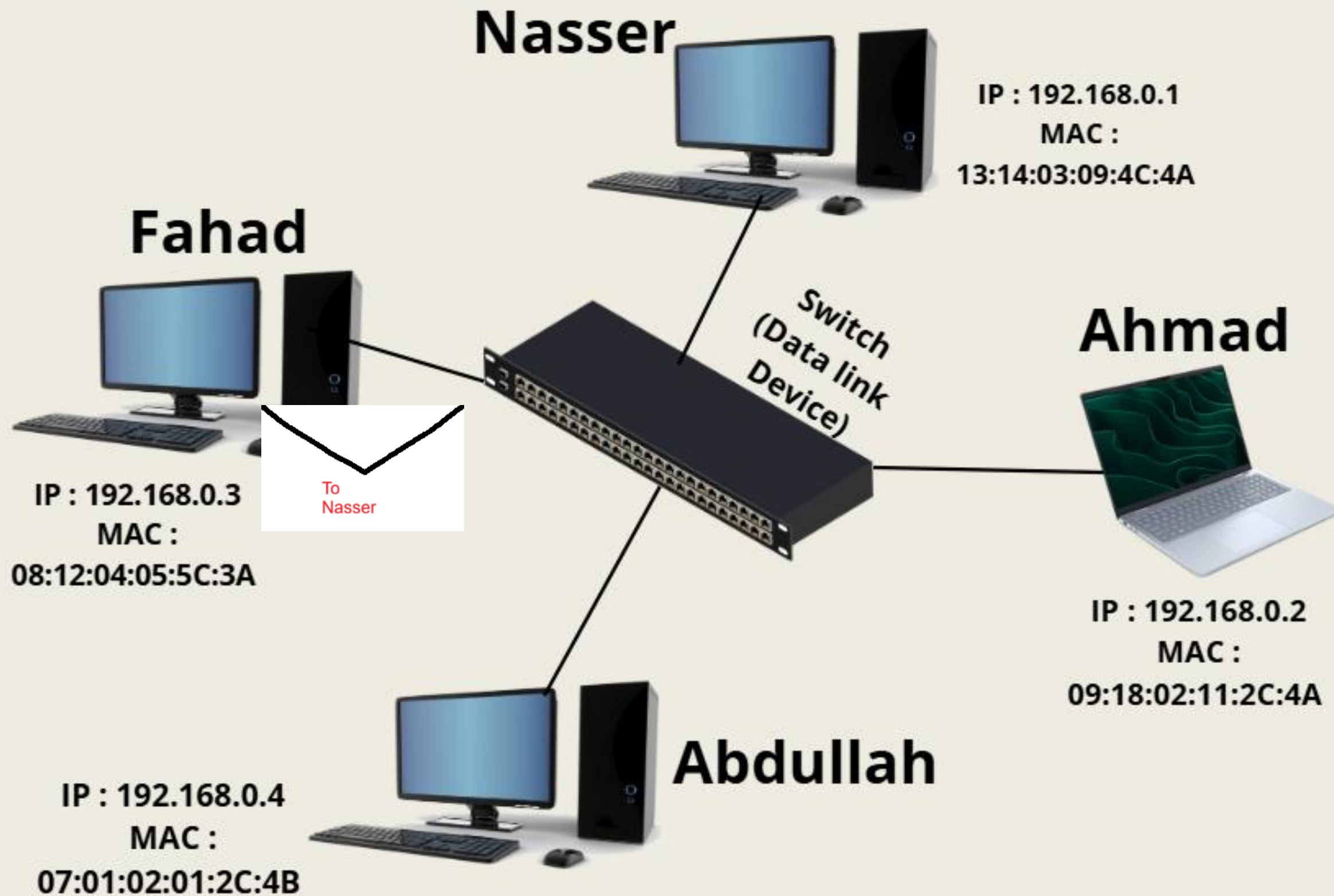
The LLC also controls the exchange of data between each two users.

Classification of MAC

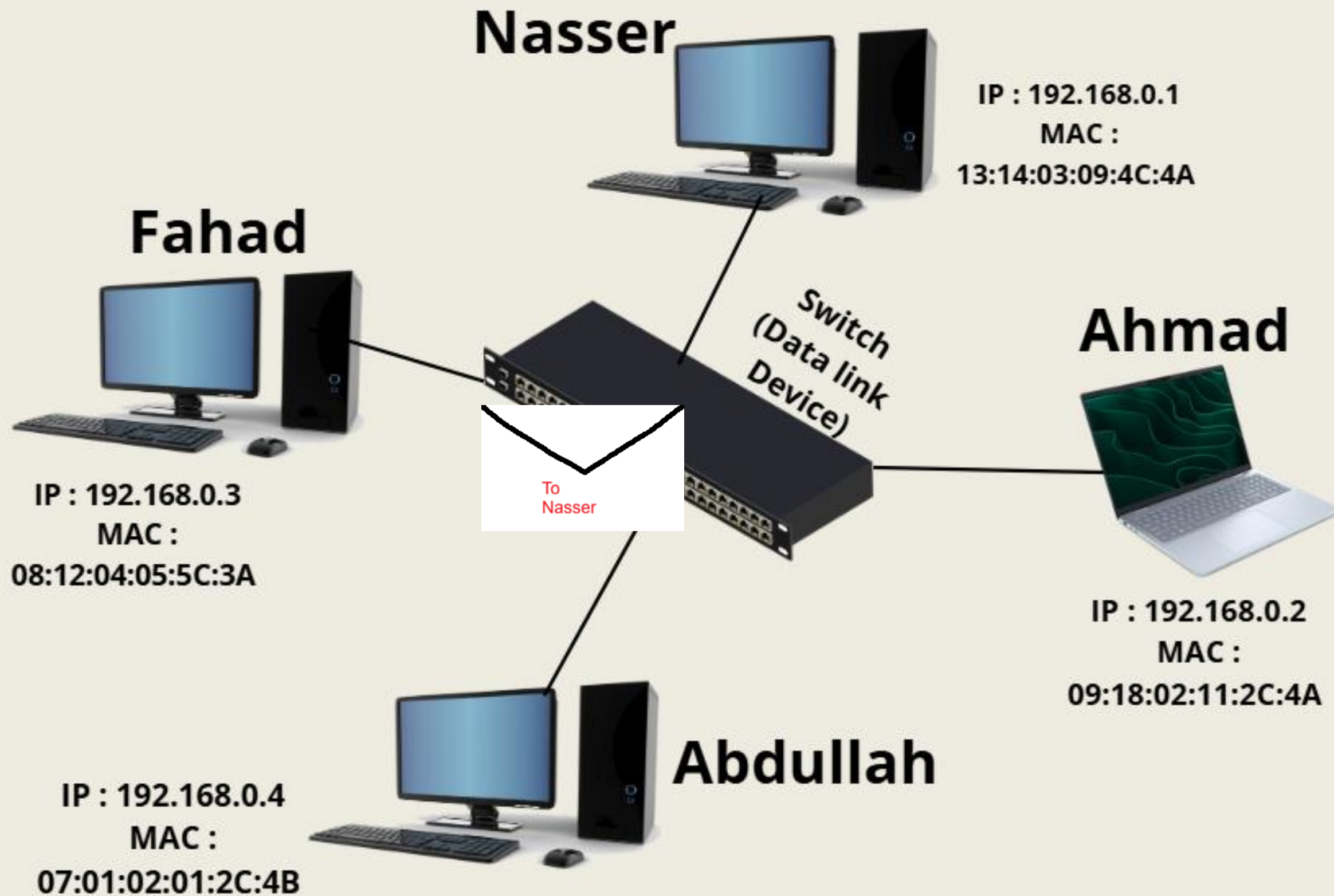
The *MAC* protocol manages access to the medium

Packets in Data-link layer (Called **Frames**) Dealing in same layer with *MAC* Address within data-link devices (Such as **Switches**)

Example



Example



Example

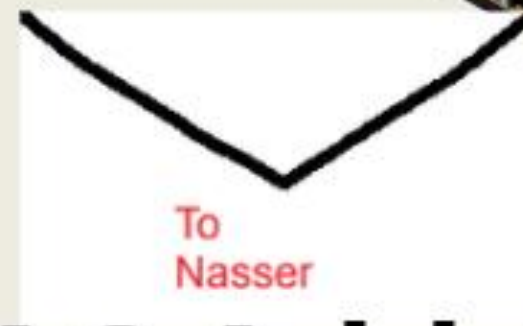
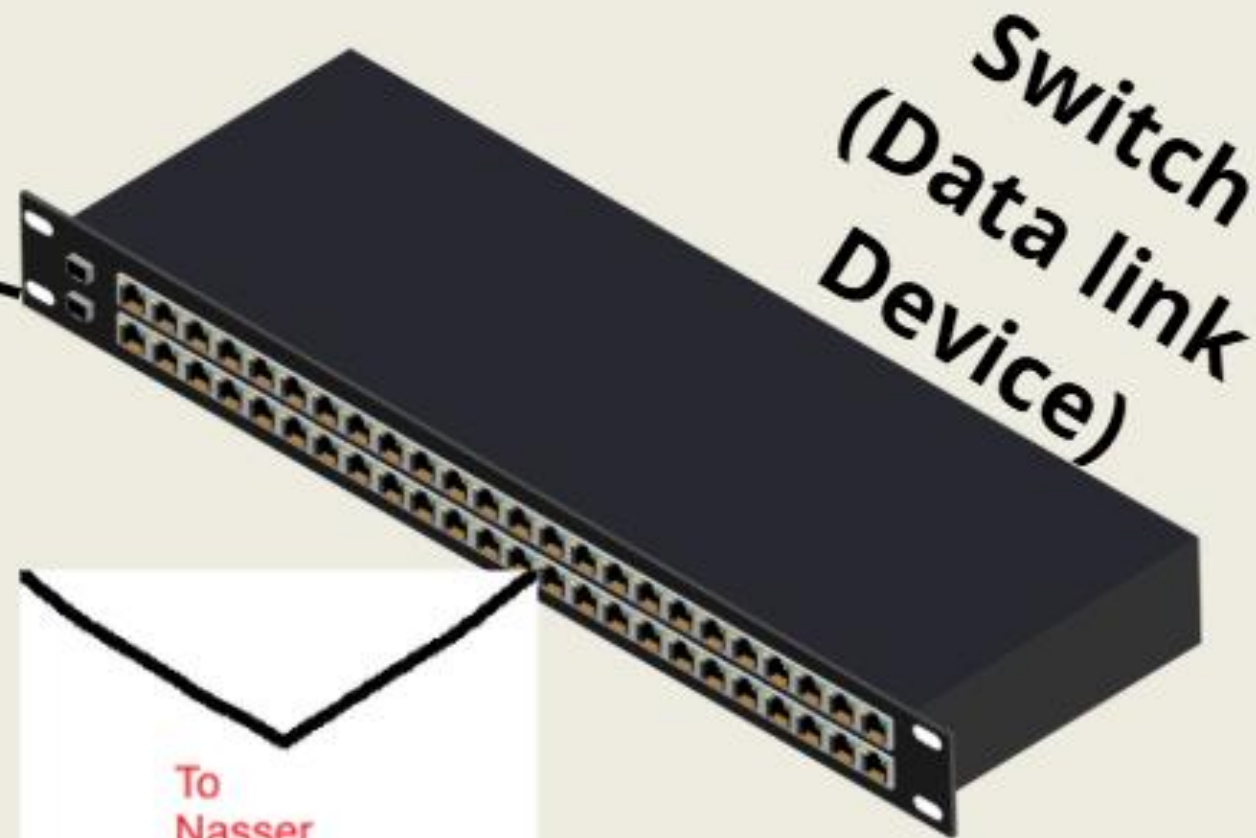
Fahad



192.168.0.3

MAC :

:04:05:5C:3A

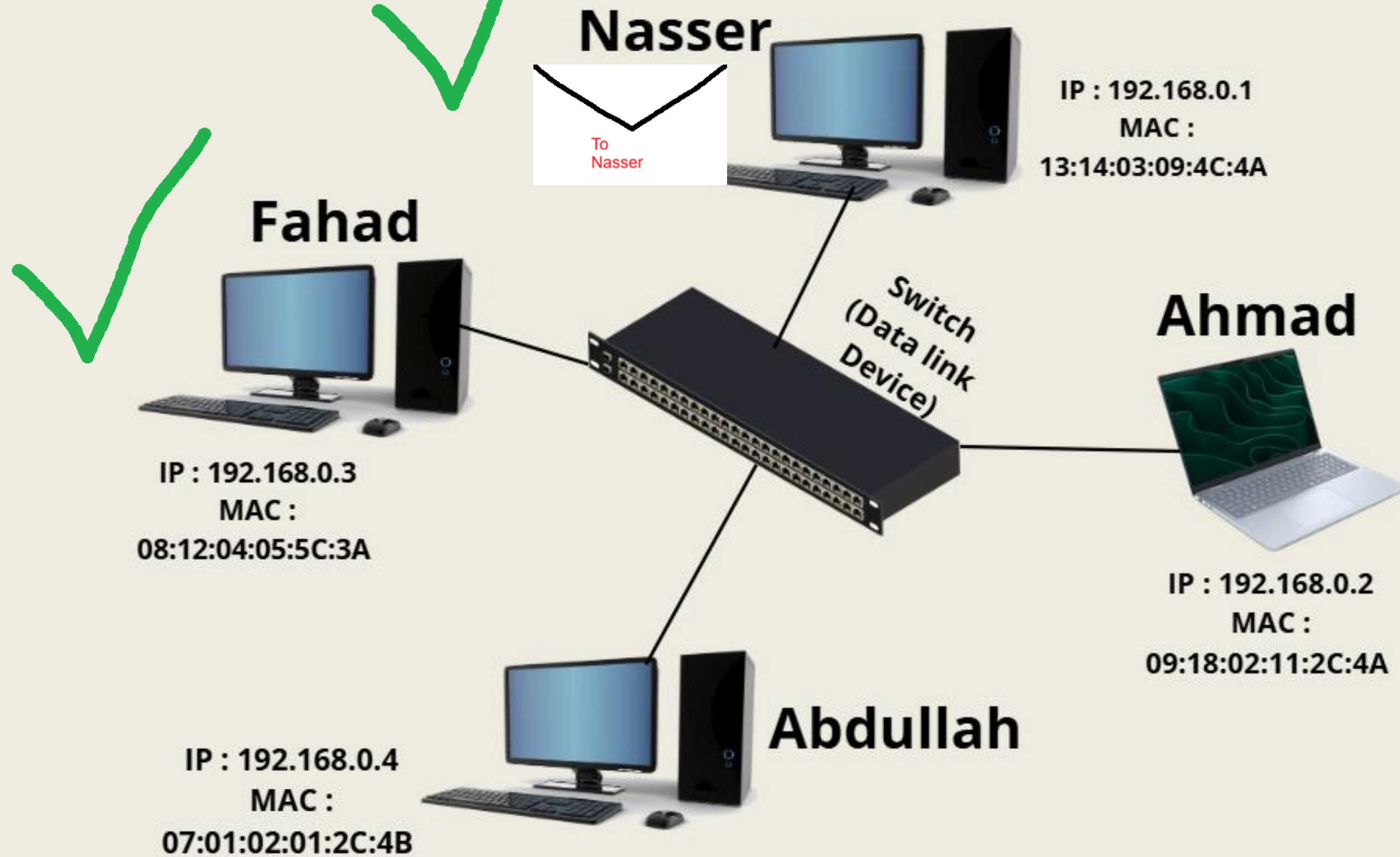


MAC Address

From 08:12:04:05:5C:3A Fahad

To 13:14:03:09:4C:4A Nasser

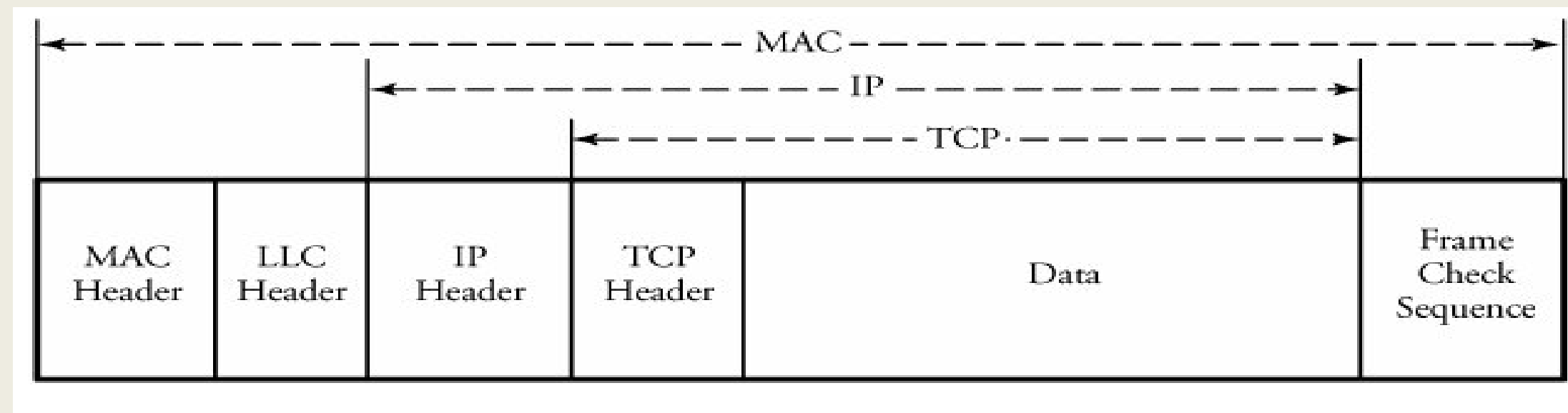
Example



Classification of MAC

The MAC protocol manages access to the medium

Packets in Data-link layer (Called **Frames**) Dealing in same layer with MAC Address within data-link devices (Such as **Switches**)



Generic MAC frame format

Classification of MAC

- MAC header gives the MAC control information, such as the MAC address of the destination and the priority level.
- LLC header contains the data from the logical-link layer.
- IP header specifies the IP header of the original packet.
- TCP header k sequence is used for error checking.

Classification of MAC

MAC protocols can be classified into two main types :

- Round Robin (RR) : Using method of famous algorithm (round robin). , more suitable for stream traffic , control is needed.
- Contention Access : needing to transmit data must contend to access the medium , No control , for light or moderate traffic.

Contention Access MAC

Carrier Sense Multiple Access (CSMA) protocols are medium access control methods that help stations avoid the collisions by listening to the channel before transmitting.

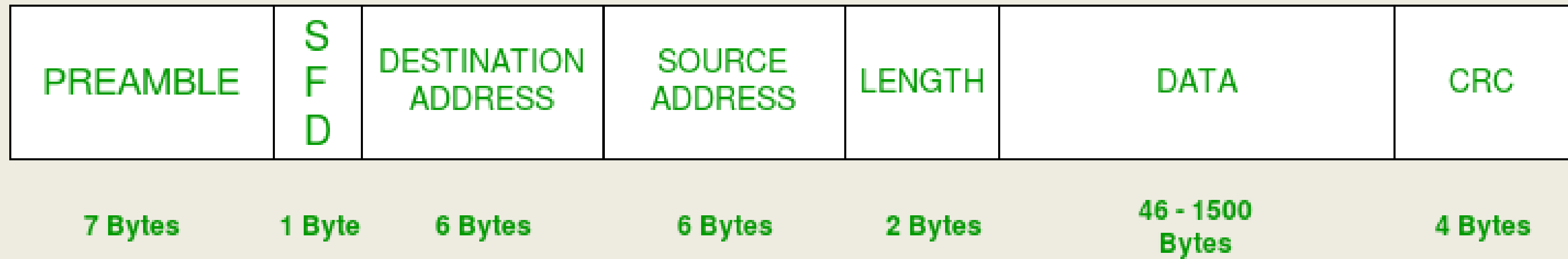
- 1 persistent CSMA — The station continuously senses the channel; when it becomes idle, it transmits immediately
- Nonpersistent CSMA — If busy, the station waits a *random* amount of time, then senses again probability (p) .
- P-persistent CSMA — Used on *slotted* channels. If idle, transmit with probability p , or defer to the next slot with probability $(1 - p)$.

Ethernet LAN: IEEE 802.3 Standard

The **IEEE 802** standards is for LAN and MAN but **IEEE 802.3** Sub-standard developed for Wired LANs called Ethernet.

The IEEE 802.3 standard uses **1-persistent CSMA** for controlling media access .

Details of IEEE 802.3 LAN frame



IEEE 802.3 ETHERNET Frame Format

Details of IEEE 802.3 LAN frame

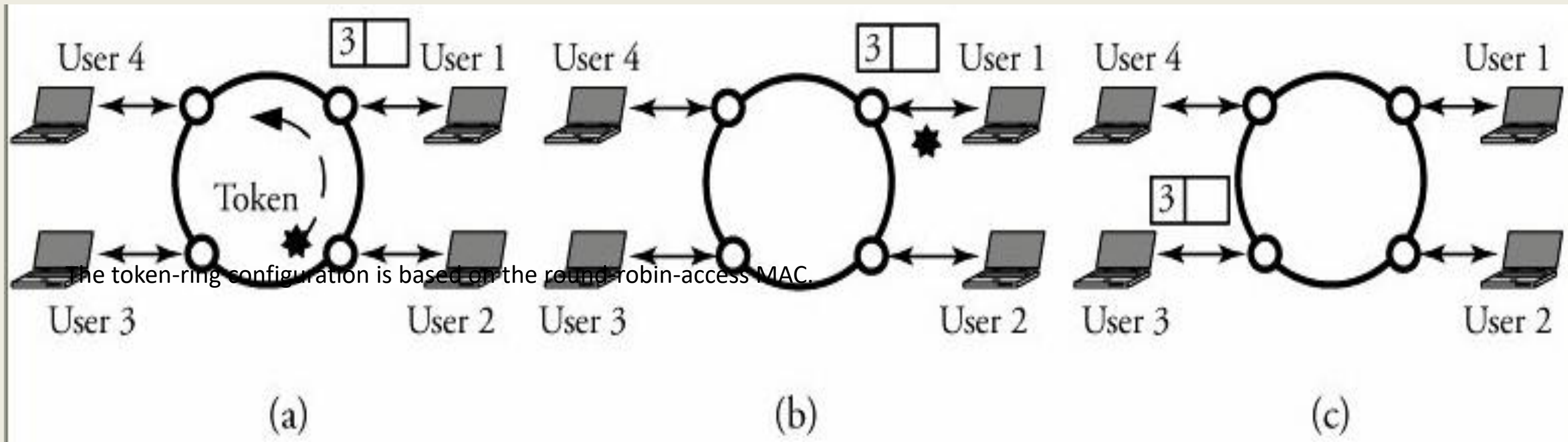
1. **PREAMBLE:** Ethernet frame starts with a 7-Bytes Preamble. This is a pattern of alternative 0's and 1's which indicates starting of the frame and allow sender and receiver to establish bit synchronization.
2. **Start of Frame :** 1-Byte field is always set to 10101011
3. **Destination Address :** 6-Byte contains the MAC address of Destination Device.
4. **Source Address :** The MAC Address of Sender
5. **Data Length**

Details of IEEE 802.3 LAN frame

6. Data : The full message sending from sender to receiver
7. CRC : or cyclic redundancy check , 4 bit contains a famous error detection method

Round Robin

Token-Ring Access Protocol or **IEEE 802.5** is based on Round Robin method , Using repeaters interconnected with unidirectional links to form a closed loop.



IEEE 802.11 WiFi Standard

IEEE 802.11 is standard used for Wireless LANs (WLAN) used WiFi Technique .

Each wireless LAN user has a wireless adapter for communication in wireless medium .

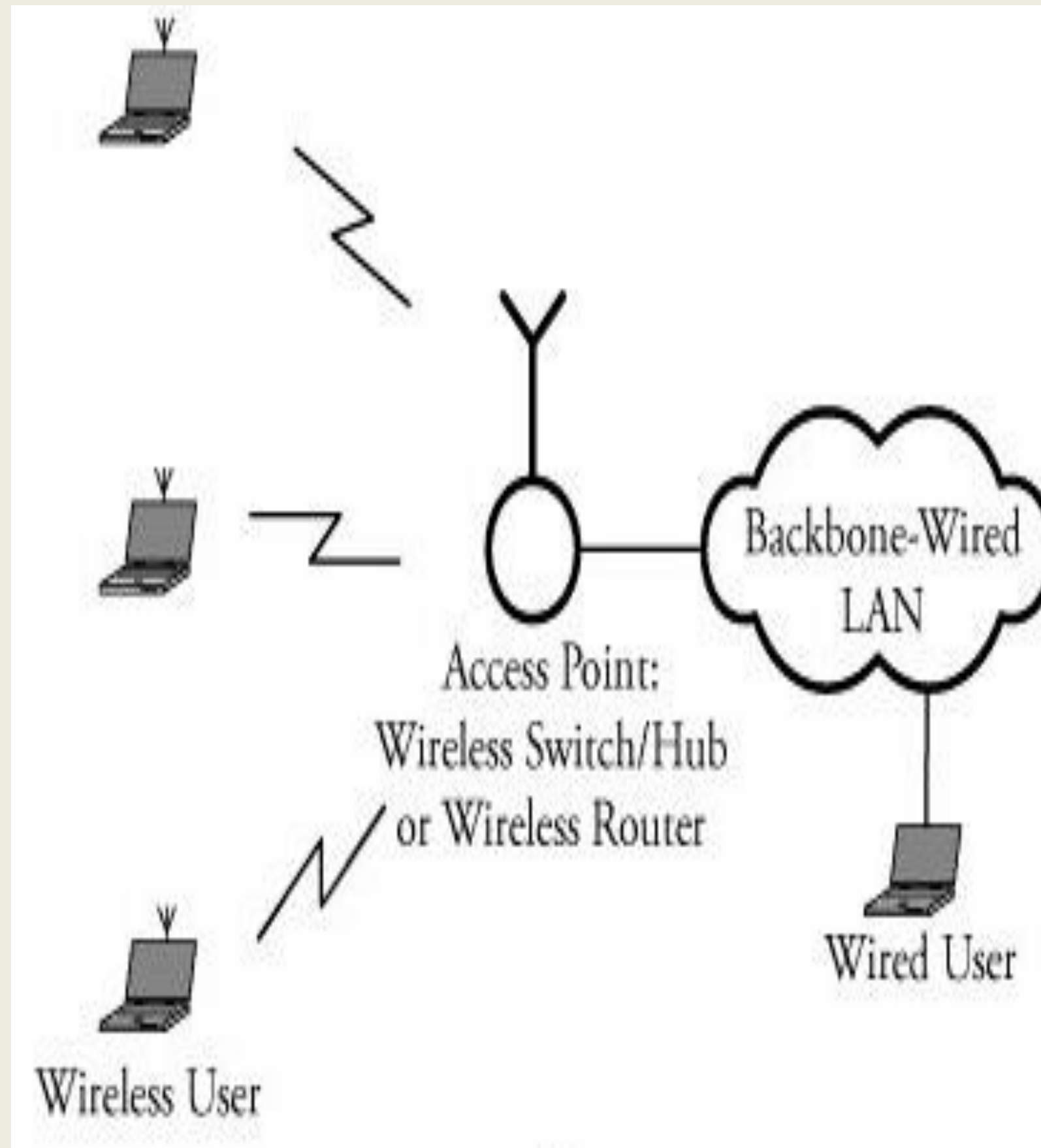
IEEE 802.11 WiFi Standard

a user in the wireless LAN first sends the data packet to the access point.

The access point recognizes the wireless user through a unique ID called the service-set identification or (**SSID**).

SSID is password-like system that enables any wireless client to join the wireless LAN used for protection WLANs .

IEEE 802.11 WiFi Standard

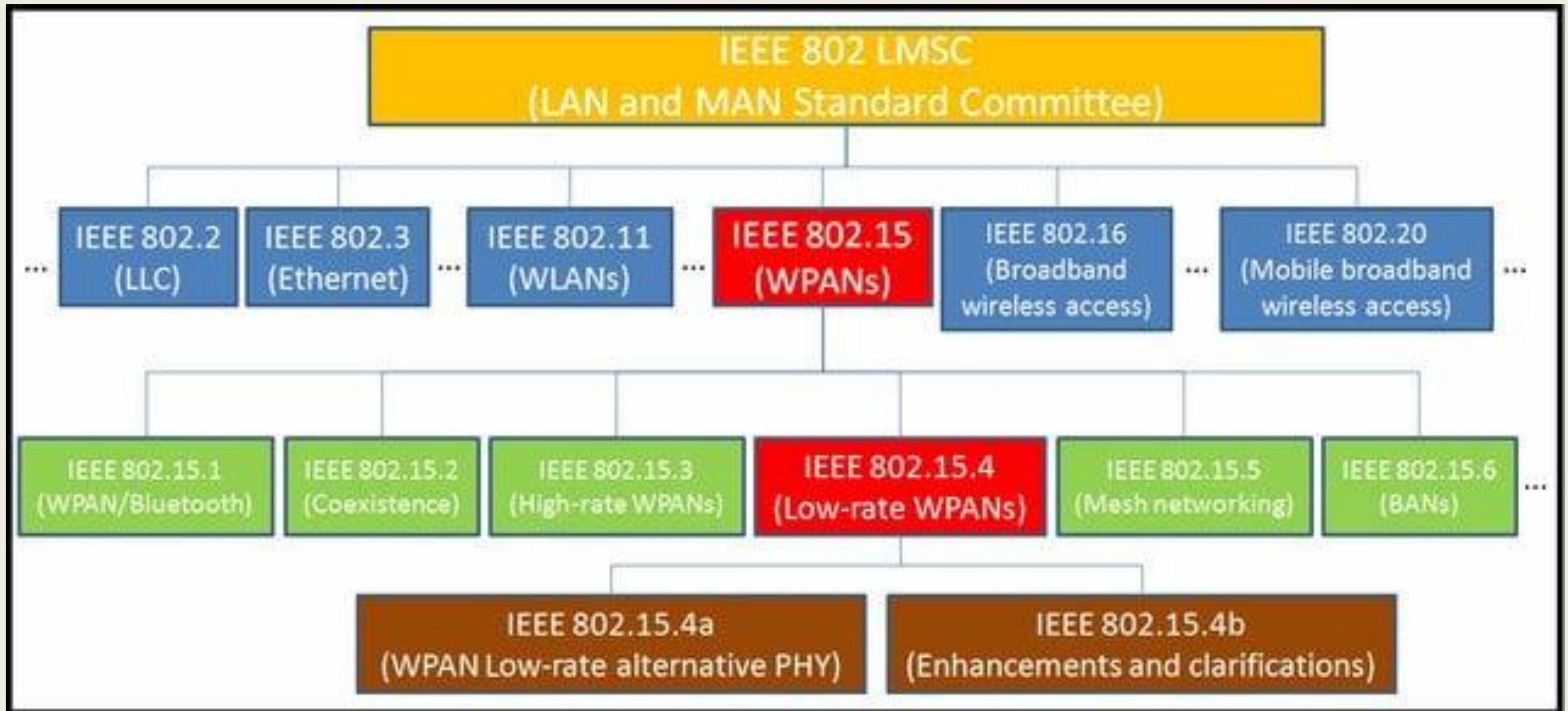


IEEE 802.11 MAC Layer

The MAC layer in IEEE 802.11 consists of two sub-layers :

- Distributed-coordination function (DCF) algorithm : mandatory MAC method, which is simpler and uses the access method known as carrier sense multiple access with collision avoidance (CSMA/CA), where stations contend for the medium
- Point-coordination function (PCF) algorithm : optional MAC method that runs on top of the DCF and is used with access points .

IEEE 802 Main Standards



Sources

- Taif University – Computer Networks ch.5 (LANs)
- Tutorialspoint – Differentiate between 1-Persistent and p-Persistent CSMA
- Data Communications and Networking By Behrouz A.Forouzan
- GeeksforGeeks – Ethernet Frame Format
- ScienceDirect – "Point Coordination Function"

Thanks for listening